

Keywords: vision-language-action models • credit assignment • robotic manipulation • GRPO

Temporal GRPO: Beyond Trajectory-Level Credit in Vision-Language-Action Reinforcement Learning

Decomposing Trajectory Rewards Into Step-Level Signals Improves Sample Efficiency and Task Success in Robotic Manipulation with VLA Policies

By Yao Zhou, Hang Gao, Fengge Wu, Changwen Zheng, Wenwen Qiang

arXiv:2608.13026 • Preprint, August 2026 • Robotics (cs.RO)

Reinforcement learning for Vision-Language-Action (VLA) models assigns a single trajectory-level reward to the entire action sequence, treating every step as equally responsible for success or failure. This coarse credit assignment conflates critical actions (precise contact) with inconsequential ones (stable approach), producing an uninformative gradient signal that slows learning. Temporal GRPO extends Group Relative Policy Optimization with a temporal credit decomposition mechanism, distributing trajectory rewards to individual steps based on their temporal context, and demonstrates improved sample efficiency and task success on robotic manipulation benchmarks.

AT A GLANCE

Problem: Trajectory-level RL credit assignment for VLA models fails to distinguish the contribution of individual actions to task outcomes.

Why it matters: VLA manipulation tasks have long action horizons where a few critical steps determine success; coarse credit wastes gradient budget.

Method: Decompose trajectory reward into step-level credits conditioned on temporal position and local state transitions within the GRPO framework.

Result: Improved sample efficiency and task success rate over standard GRPO on multi-step robotic manipulation benchmarks.

Evidence: Gains replicated across multiple manipulation tasks; ablations confirm temporal conditioning is the key driver.

The Big Picture

Vision-Language-Action (VLA) models are large multi-modal policies that take images of the scene and a language instruction as input and output a sequence of robot actions. They have achieved impressive results on diverse

manipulation tasks by combining the knowledge of large pretrained vision-language models with task-specific RL fine-tuning.

The RL fine-tuning phase typically uses algorithms like GRPO, which generates groups of G trajectories from the current policy, computes a binary reward (did the task succeed?), and uses the relative ranking within the group to compute policy gradient updates. Every action in a successful trajectory receives a positive update; every action in a failed trajectory receives a negative update.

This trajectory-level assignment is coarse. A grasping task with 80 action steps may succeed or fail based on a single step — the precise contact between the gripper and the object. Temporal GRPO identifies that step and concentrates gradient signal on it, leaving the uninformative steps with near-zero gradient.

Why Existing Approaches Fall Short

Standard GRPO assigns trajectory reward uniformly across all steps:

$$\nabla \mathcal{L} = \frac{1}{G} \sum_{g=1}^G \hat{A}_g \cdot \sum_{t=1}^T \nabla \log \pi_{\theta}(a_{g,t} | s_{g,t})$$

where $\hat{A}_g = R_g - \text{baseline}$ is the group-relative advantage. Every step t in trajectory g receives the same weight $\hat{A}_g$.

Dense reward shaping assigns step-level rewards via a hand-designed function (e.g., distance to goal, object height). This works for simple tasks but requires domain expertise and does not generalize across task families.

Hindsight reward assignment assigns credit based on outcome retrospectively but does not distinguish which actions were causally important for that outcome.

Temporal GRPO avoids hand-designed rewards and operates within the standard GRPO framework, requiring only a temporal credit decomposition function that is learned from the trajectory data.

Core Mechanism

Temporal GRPO decomposes the trajectory-level advantage $\hat{A}_g$ into step-level advantages $\hat{a}_{g,t}$ using a temporal credit function $c(t, T, s_{g,t})$ that depends on:

- **Temporal position t/T :** Early steps (approach) and late steps (contact, post-grasp) receive different baseline credit weights.
- **State transition:** The change in scene state $\Delta s_{g,t} = s_{g,t+1} - s_{g,t}$ identifies steps where significant state changes occurred (contact, placement). High-change steps receive higher credit.

The group-relative baseline is computed at the step level: for step t across all G trajectories, the baseline is the mean

step-level credit over the group:

$$\hat{a}_{g,t} = c(t, T, s_{g,t}) \cdot \hat{A}_g - \bar{c}_t$$

where $\bar{c}_t = \frac{1}{G} \sum_{g'=1}^G c(t, T, s_{g',t}) \cdot \hat{A}_{g'}$.

Technical Formulation

The Temporal GRPO policy gradient objective is:

$$\mathcal{L}_{\text{TG}}(\theta) = -\frac{1}{G} \sum_{g=1}^G \sum_{t=1}^T \hat{a}_{g,t} \cdot \log \pi_{\theta}(a_{g,t} \mid s_{g,t})$$

with credit function:

$$c(t, T, s_{g,t}) = w_{\text{pos}}(t/T) \cdot w_{\text{trans}}(\|\Delta s_{g,t}\|)$$

where w_{pos} is a learned position-dependent weight function (modeled as a shallow MLP) and w_{trans} is a monotone function of state transition magnitude.

The position weight w_{pos} is trained jointly with π_{θ} via a meta-learning objective that maximizes the correlation between step-level advantage and final task success probability:

$$\mathcal{L}_{\text{meta}}(w) = -\text{Corr}(\hat{a}_{g,t}, P(\text{success} \mid s_{g,1:t}))$$

where $P(\text{success} \mid s_{g,1:t})$ is estimated by a lightweight success predictor trained on the same trajectory data.

Learning or Inference Procedure

Training alternates between two updates:

1. **Policy update:** Sample G trajectories from π_{θ} , compute $\hat{a}_{g,t}$ using current w_{pos} , update θ via $\mathcal{L}_{\text{TG}}$.
2. **Credit function update:** Update w_{pos} via $\mathcal{L}_{\text{meta}}$ using the current trajectory data.

The success predictor is updated with every new trajectory batch. At inference, the VLA policy π_{θ} is used directly; no credit function is needed.

What the Guarantee Says

No formal guarantee is provided. The empirical claim is that temporal credit decomposition consistently improves sample efficiency (fewer trajectories to reach a target success rate) and asymptotic task success rate over standard GRPO, across the tested manipulation tasks. The meta-learning correlation objective provides a principled basis for learning the credit function, but convergence properties are empirical.

Experimental Findings

- **Sample efficiency:** Temporal GRPO reaches 80% task success in 40–55% fewer environment interactions

than standard GRPO across grasping, placing, and stacking tasks.

- **Asymptotic success:** Final success rates improve by 5–12 percentage points on complex long-horizon tasks (4+ subtasks).
- **Long-horizon advantage:** Improvement scales with task horizon. Short tasks (1–2 steps) show minimal gain; tasks with 6+ action phases show the largest improvements.
- **Contact step analysis:** The credit function concentrates >60% of the total trajectory credit on the 2–5 contact steps in a typical 80-step grasp trajectory, confirming that temporal credit is identifying the causally important steps.

Ablations and Interpretation

Position weighting only (removing state-transition weighting) captures 60–70% of the full benefit, confirming that temporal position is the dominant signal. The state-transition term adds the remainder.

Fixed vs. learned position weight: A hand-designed position weight (e.g., linear ramp emphasizing later steps) achieves 80% of the benefit of the learned weight on structured manipulation tasks, suggesting the timing pattern is partially predictable from task structure.

Group size G : Temporal GRPO achieves the same accuracy as standard GRPO with $G = 8$ using $G = 4$, halving the rollout cost for the same gradient quality.

Reference

Yao Zhou, Hang Gao, Fengge Wu, Changwen Zheng, Wenwen Qiang. **Temporal GRPO: Beyond Trajectory-Level Credit in Vision-Language-Action Reinforcement Learning**. arXiv:2608.13026, August 2026. <https://arxiv.org/abs/2608.13026>